

Global Model Parallel Sequencing - v17+

Updated Workflow Analysis (February 2026)

Based on global-workflow commit 2a679b4a | 89 J-jobs | 6 Application Types

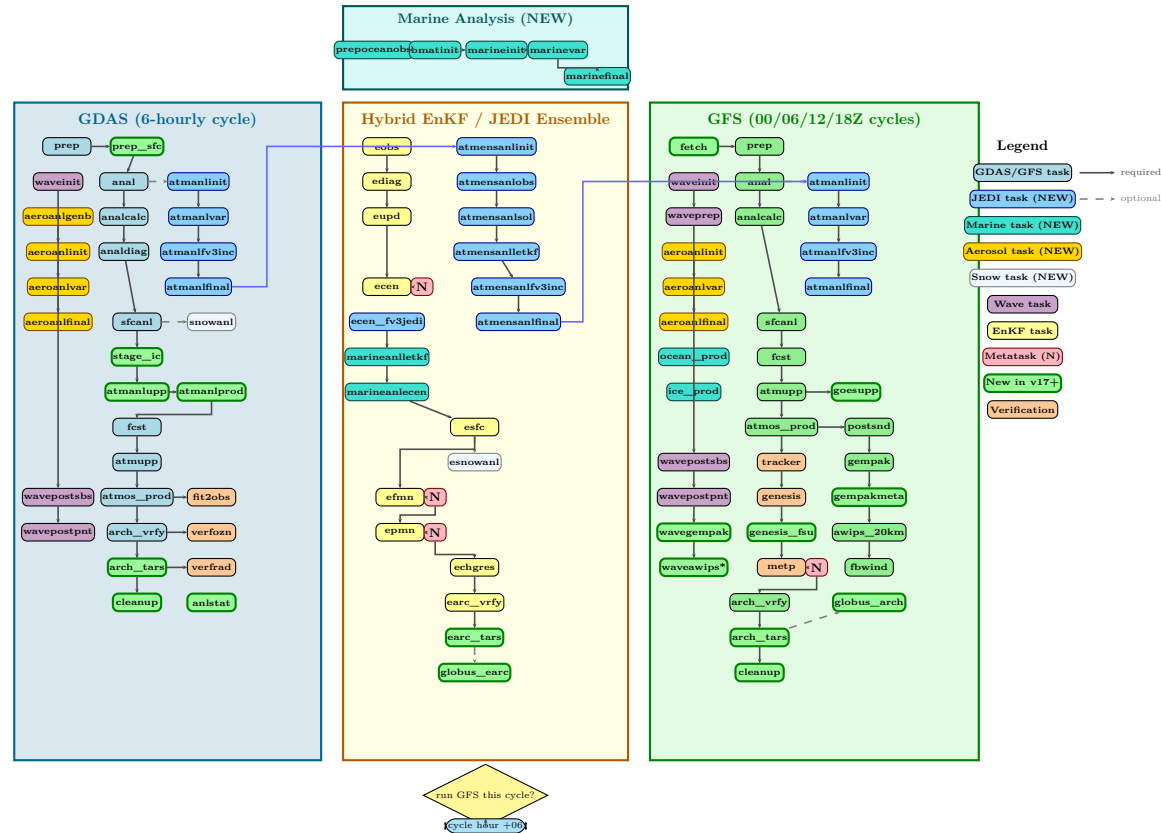


Figure 1: Global Model Parallel Sequencing v17+ — Updated from v16 with JEDI, Marine, and Aerosol Analysis Systems

Changes from v16 to v17+

Table 1: New Tasks Added Since v16

Category	Task(s)	Description
JEDI Atmospheric Analysis (replaces GSI optionally)		
	atmanlinit	Initialize JEDI atm analysis
	atmanlvar	JEDI variational solver
	atmanlfv3inc	Generate FV3 increment
	atmanlfinal	Finalize JEDI analysis
	analcac_fv3jedi	FV3JEDI analysis calculation
JEDI Ensemble Analysis		
	atmensanlinit	Initialize ensemble analysis
	atmensanlobs	Ensemble observation processing
	atmensanlsol	Ensemble solver
	atmensanlletkf	LETKF solver
	atmensanlfv3inc	Ensemble FV3 increment
	atmensanlfinal	Finalize ensemble analysis
	ecen_fv3jedi	FV3JEDI ensemble centering
Marine Analysis System (SOCA)		
	prepoceanobs	Prepare ocean observations
	marinebmatinit	Initialize marine B-matrix
	marinebmat	Compute marine B-matrix
	marineanlinit	Initialize marine analysis
	marineanlvar	Marine variational analysis
	marineanlchkpt	Marine analysis checkpoint
	marineanlfinal	Finalize marine analysis
	marineanlletkf	Marine LETKF (ensemble)
	marineanlece	Marine ensemble centering
Aerosol Analysis		
	aeroanlgenb	Generate aerosol B-matrix
	aeroanlinit	Initialize aerosol analysis
	aeroanlvar	Aerosol variational analysis
	aeroanlfinal	Finalize aerosol analysis
	aerosol_init	Aerosol initialization (fcst-only)
Snow Analysis		
	snowanl	JEDI snow analysis
	esnowanl	Ensemble snow analysis
New Products		
	ocean_prod	Ocean products
	ice_prod	Ice products
	atmanlupp	Analysis UPP
	atmanlprod	Analysis products

Table 2: Tasks Removed or Renamed from v16

v16 Task	Status	Notes
gldas	Removed	GLDAS land DA deprecated
post/postN	Renamed	Now atmupp + atmos_prod
arch	Split	Now arch_vrfy + arch_tars
earc/eamn	Renamed	Now earc_vrfy + earc_tars
vrfy	Expanded	Multiple verification tasks

New Application Types Since v16

- **SFS** (Subseasonal Forecast System) — New forecast-only application
- **GCAFS** (Global Climate and Air quality Forecasting System) — Cycled and forecast-only modes
- **GEFS** (Global Ensemble Forecast System) — Enhanced with 21 tasks

Task Count Comparison

Metric	v16	v17+
Total J-Jobs	~50	89
GFS Tasks (gfs_tasks.py)	~35	89 methods
GEFS Tasks (gefs_tasks.py)	—	21
SFS Tasks (sfs_tasks.py)	—	18
Application Types	2	6